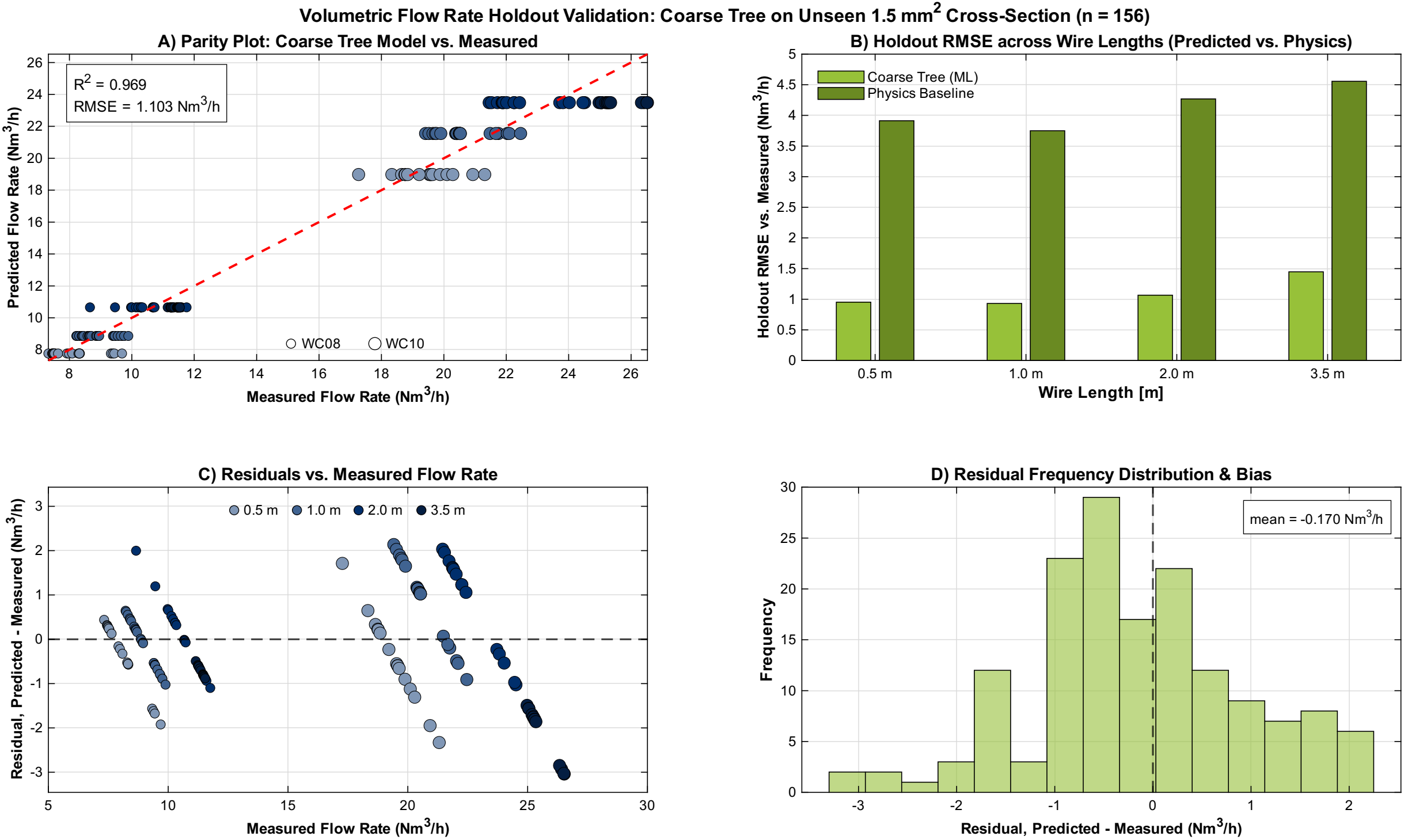

Volumetric Flow Rate Holdout Validation: Coarse Tree on Unseen 1.5 mm² Cross-Section (n = 156)
A) Parity Plot: Coarse Tree Model vs. Measured
R² = 0.969
RMSE = 1.103 Nm³/h
WC08
WC10
Predicted Flow Rate (Nm³/h)
Measured Flow Rate (Nm³/h)
B) Holdout RMSE across Wire Lengths (Predicted vs. Physics)
Coarse Tree (ML)
Physics Baseline
Holdout RMSE vs. Measured (Nm³/h)
0.5 m
1.0 m
2.0 m
3.5 m
Wire Length [m]
C) Residuals vs. Measured Flow Rate
0.5 m
1.0 m
2.0 m
3.5 m
Residual, Predicted - Measured (Nm³/h)
Measured Flow Rate (Nm³/h)
D) Residual Frequency Distribution & Bias
mean = -0.170 Nm³/h
Frequency
Residual, Predicted - Measured (Nm³/h)